Transplantation Antigens and Graft-Versus-Host Disease (GVHD)

Transplantation antigens trigger immune responses leading to the targeting of host tissues, particularly epithelial tissues, for cytotoxic damage. The precise mechanism of lymphocyte-mediated injury remains unclear—whether it occurs primarily through direct cytotoxic molecules such as **granzyme** and **perforin**, through cytokine-mediated pathways, or via a combination of both. Epithelial cell death proceeds predominantly by **apoptosis**, with evidence supporting roles for both **T cell-mediated mechanisms** and **noncellular triggers**.

There is no consistent involvement of a single lymphocyte subset in mediating GVHD. Multiple epitopes on major and minor histocompatibility antigens can drive allostimulation, resulting in diverse populations of effector lymphocytes within target organs. For example, several cytotoxic **T-cell clones** have been isolated from the skin of a single patient. Both **$CD4^+$** and **$CD8^+$** T cells are capable of inducing epidermal damage. In murine models, the effector T-cell population responsible for cutaneous GVHD varies depending on the transplant strain combination, suggesting that each human donor-recipient pair may elicit a unique immune response involving different lymphocyte subsets. Nonetheless, clinical evidence supports a central role for **$CD4^+$ T cells** in human GVHD pathogenesis. This immunologic diversity likely contributes to the variable clinical expression of GVHD, independent of MHC matching status.

Other immune cells also participate in GVHD. The presence of abundant **natural killer (NK) cells** in dermal and epidermal infiltrates during cutaneous GVHD suggests their recruitment following antigen-specific T-cell activation in target tissues. Additionally, increases in **mast cells** and **dermal dendrocytes** are observed in the superficial dermis during acute disease phases.

The reason for the frequent and early involvement of the skin in GVHD compared to liver or gastrointestinal manifestations remains unknown.

Chronic Cutaneous Graft-Versus-Host Reaction after Allogeneic Stem Cell Transplantation

Chronic cutaneous GVHD closely resembles autoimmune disorders such as **lichen planus**, **scleroderma**, and **Sjögren's syndrome**, suggesting shared pathogenic mechanisms. Dysregulation of autoreactive immune components is thought to underlie these conditions. **Transforming growth factor-β (TGF-β)** plays a key role in the development of fibrotic skin and visceral changes seen in sclerodermoid GVHD, with studies showing that neutralizing antibodies against TGF-β can reverse fibrosis in mouse models.

Notably, in women with scleroderma, persistent **male fetal microchimerism** has been detected, prompting the hypothesis that a fetal anti-maternal immune reaction—akin to GVHD—may contribute to disease pathogenesis. In sclerodermatous skin, fibroblasts aberrantly express **MHC class II molecules** and produce excessive amounts of **type I collagen**, further supporting an immune-driven fibrotic process.

With the increasing use of **peripheral blood stem cell transplantation**, there may be a higher incidence of chronic GVHD, a consideration important for long-term patient management.

Acute Cutaneous Graft-Versus-Host Reaction after Autologous and Syngeneic Marrow and Peripheral Stem Cell Transplantation

Evidence from rodent and human studies indicates that autoimmunity may develop against epitopes on **MHC class II molecules**. Since **IFN-γ** upregulates MHC class II expression—particularly on tumor cells—and **cyclosporine** promotes the expansion of MHC class II-reactive T cells, the combination of these factors with high-dose chemotherapy in autologous transplantation may enhance **GVH-like cutaneous reactions**. This phenomenon might also contribute to a beneficial **graft-versus-tumor effect**, even in the absence of allogeneic immunity.

▲FIGURE 28-1 Acute cutaneous graft-versus-host reaction (GVHR). Erythematous macules involving the pinna (A), palms (B), and soles are typical in early stages of cutaneous GVHR after allogeneic and autologous marrow transplantation.

CLINICAL FINDINGS

History

A thorough patient history is essential in evaluating suspected GVHD. Key elements include:
- Type of transplant (allogeneic, autologous, syngeneic)
- Time elapsed since transplantation

- Conditioning regimen used
- Use of immunosuppressive agents (e.g., cyclosporine, tacrolimus)
- Presence of prior acute GVHD
- Donor-recipient HLA match status
- Recent infections or drug exposures

These factors help differentiate GVHD from other dermatoses and guide further diagnostic evaluation.